

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
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S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1002+200

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

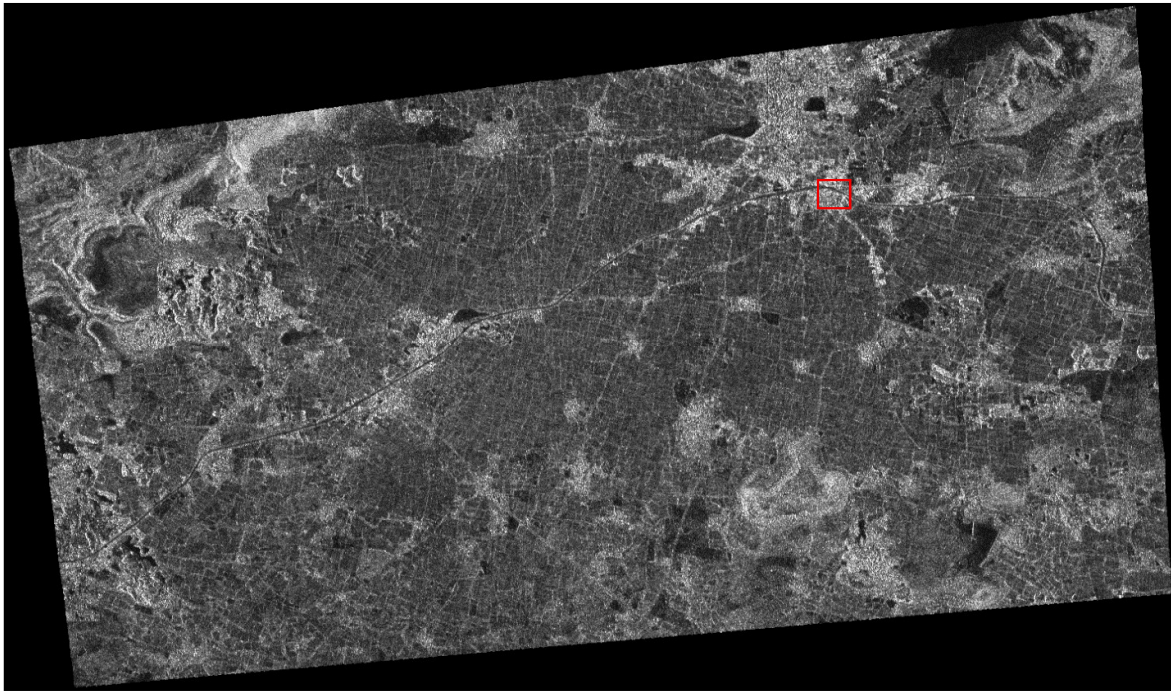
Flight Direction - DESCENDING

Polarization – VV+VH

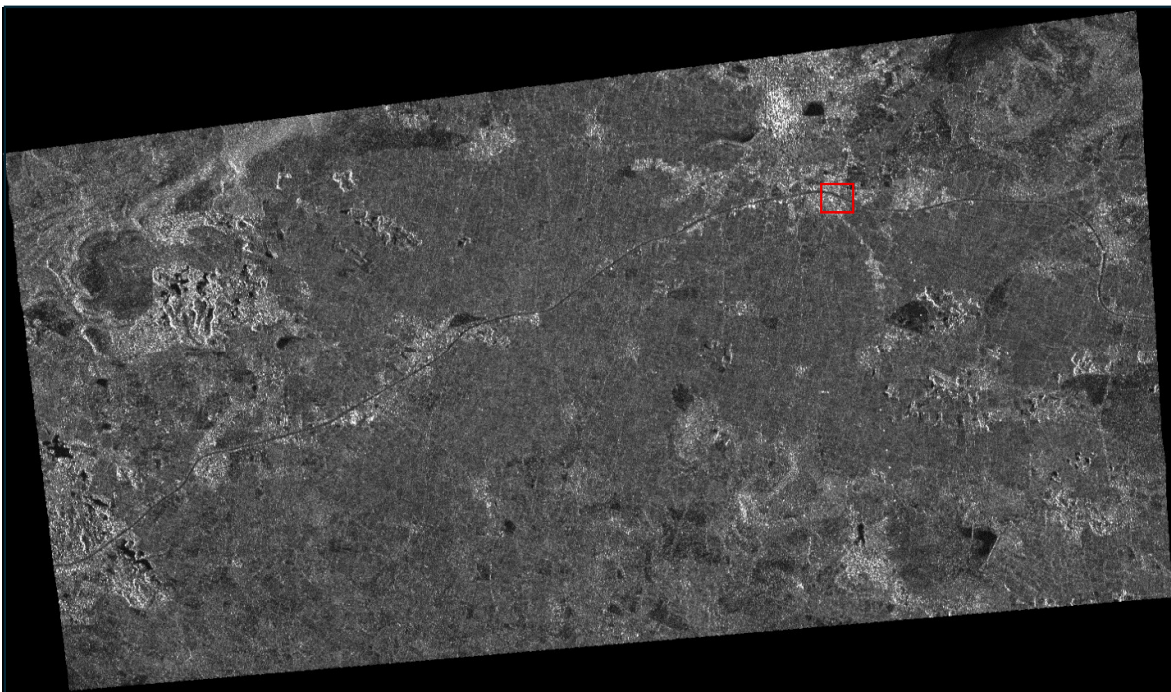
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

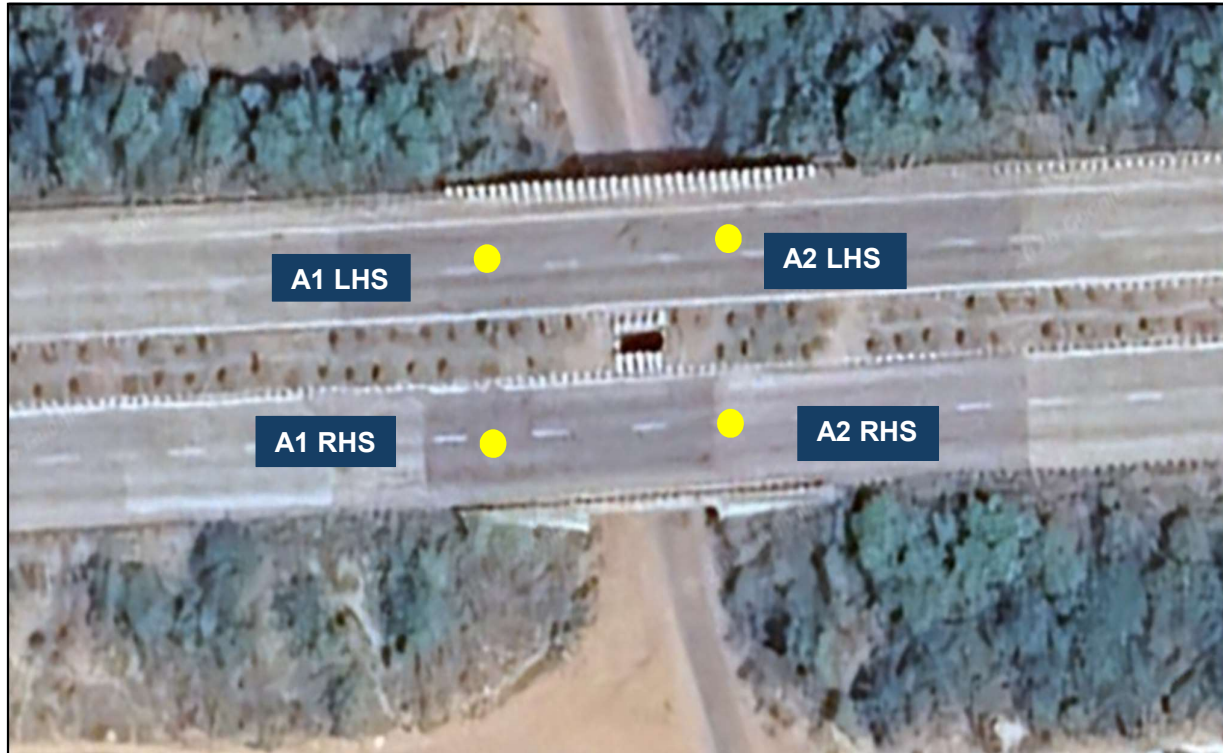


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

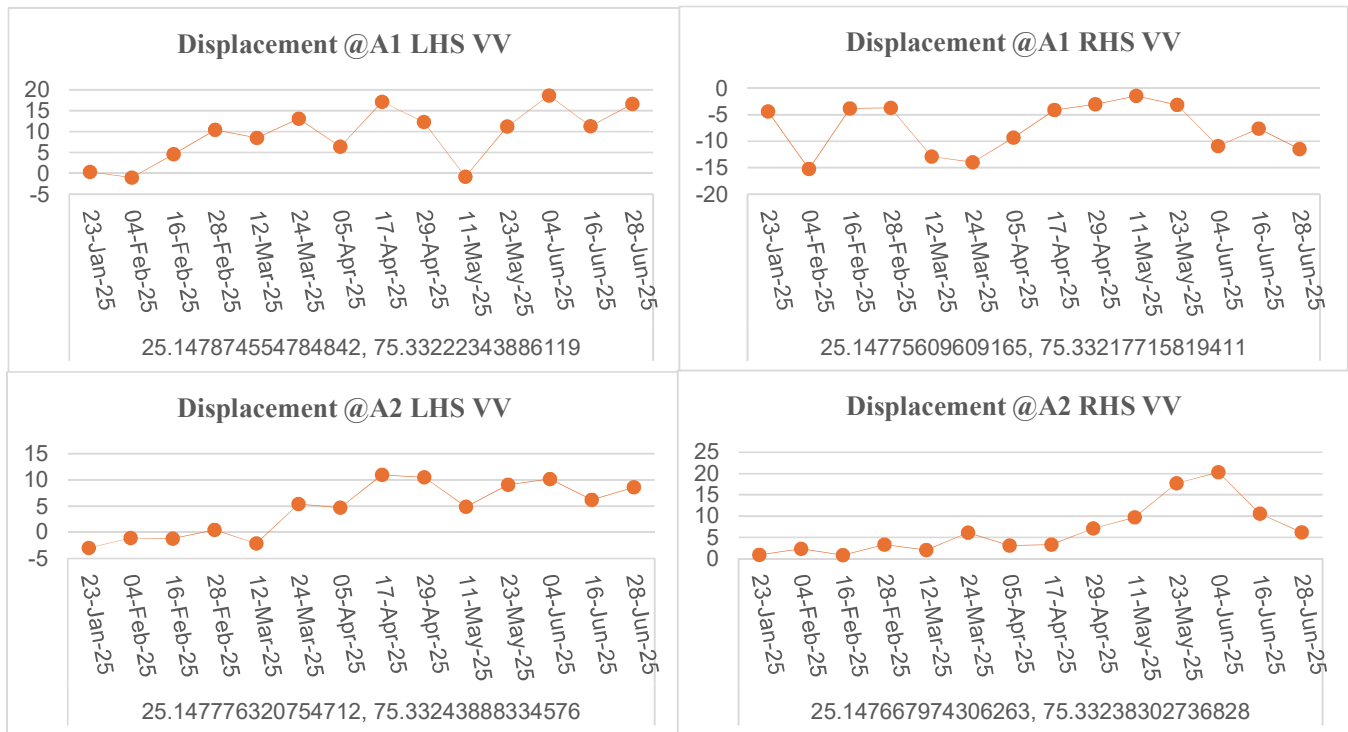
VUP – CH-1002+200

Length – 33.34M

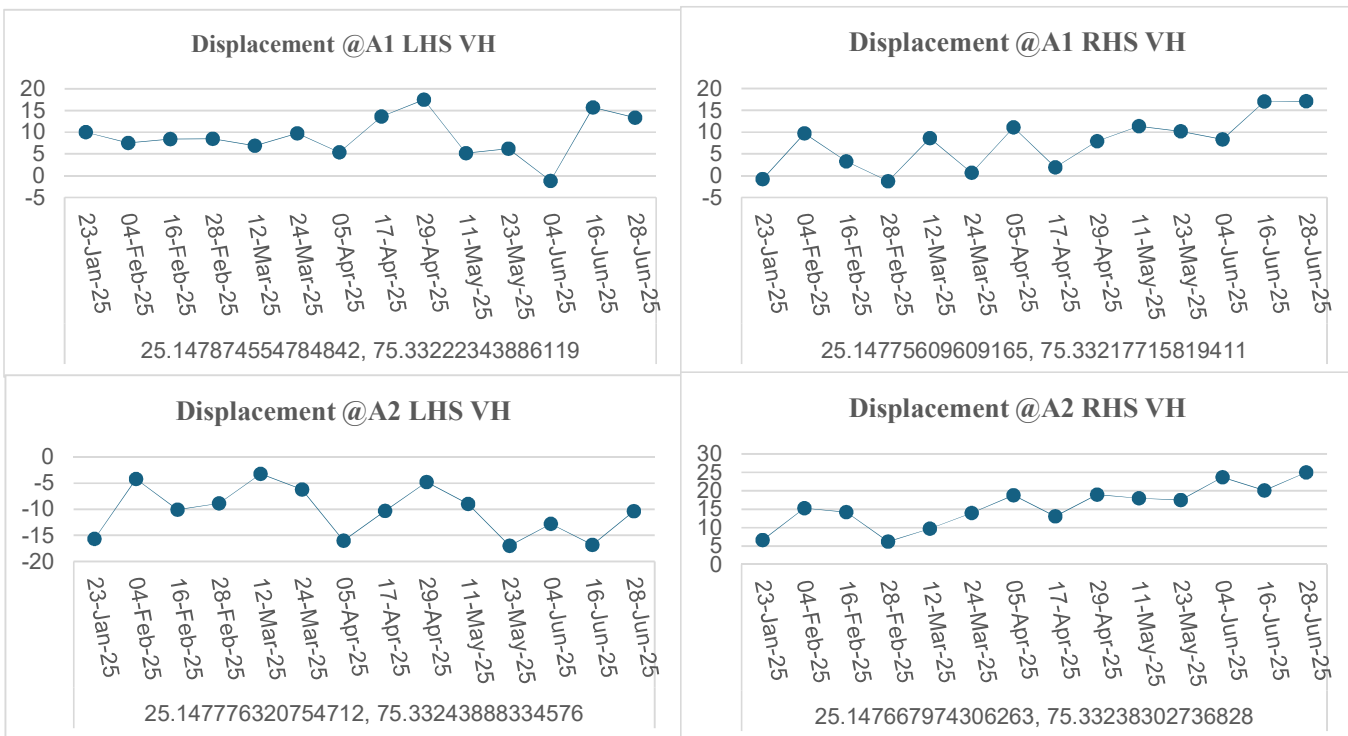


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



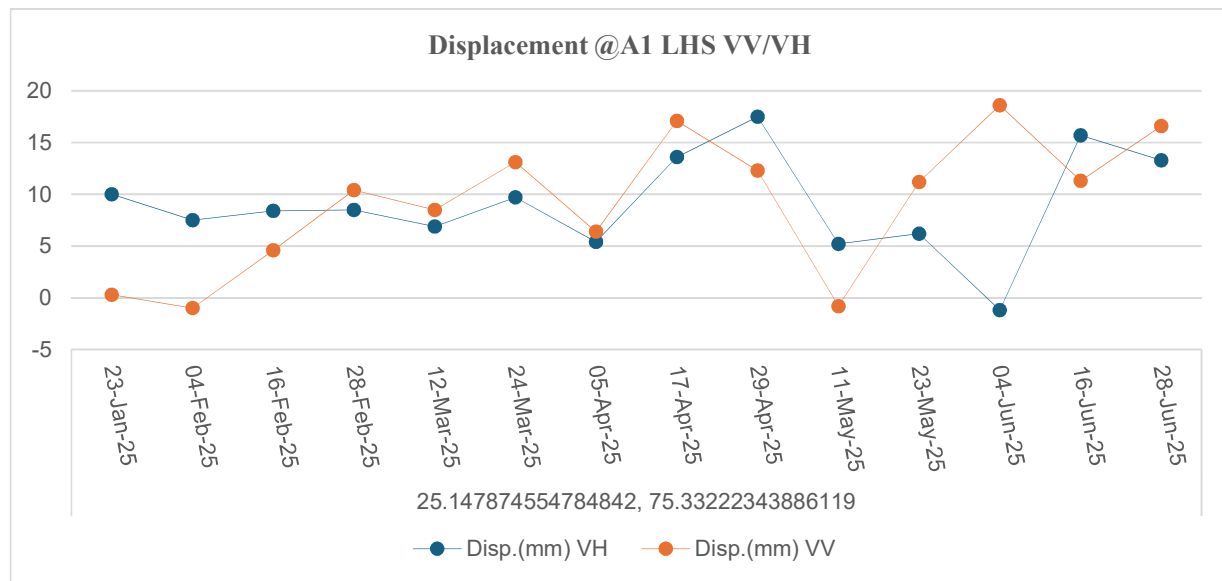
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.147874554784842, 75.33222343886119)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows a predominantly positive trend, varying roughly from 0 mm to +16 mm.
- **Horizontal Displacement (VH)** exhibits a moderate fluctuating trend, ranging approximately from -10 mm to +12 mm.

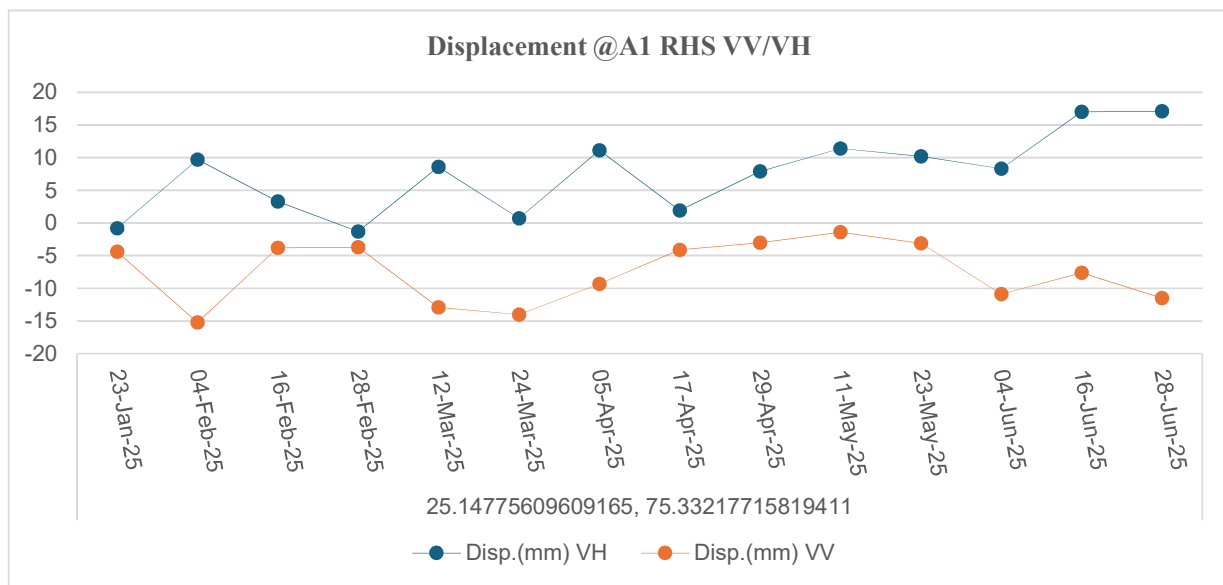
Critical Events & Anomalies:

- **January 2025:** VH remains slightly positive (+2 mm) while VV is relatively high (+14 mm), indicating initial vertical uplift with minimal lateral movement.
- **May-June 2025:** Both VH and VV stabilize at moderate positive values (VH +5 to +12 mm, VV +6 to +12 mm), suggesting a balanced and stable displacement regime

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.14775609609165, 75.33217715819411)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows an overall positive and increasing trend, ranging approximately from –2 mm to +17 mm.
- **Vertical displacement (VV)** Vertical displacement remains consistently negative, varying roughly between –15 mm and –2 mm.

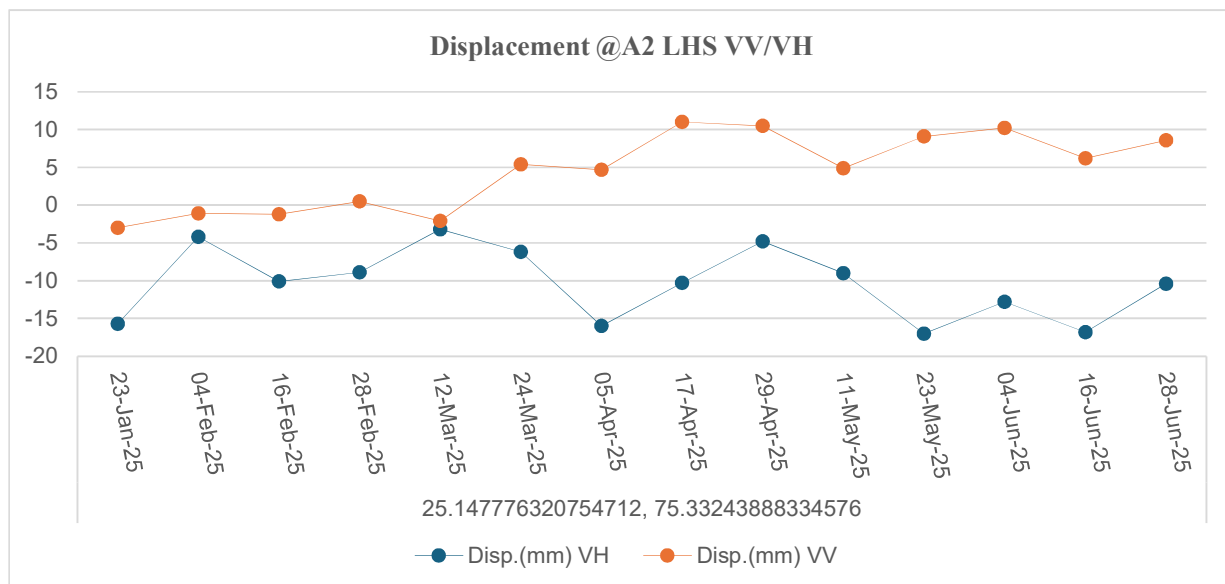
Critical Events & Anomalies:

- **February 2025:** VH reaches an early positive peak (+10 mm) while VV records a sharp settlement (–15 mm), suggesting significant vertical compression accompanied by lateral response.
- **June 2025:** VH reaches its maximum levels (+16 to +17 mm) while VV deepens again (–10 to –12 mm), indicating long-term lateral progression coupled with renewed vertical settlement.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.147776320754712, 75.33243888334576)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** is predominantly negative throughout the monitoring period, ranging approximately from -4 mm to -18 mm.
- **Vertical Displacement (VV)** transitions from slightly negative values (-3 to -1 mm) in the early phase to consistently positive values.

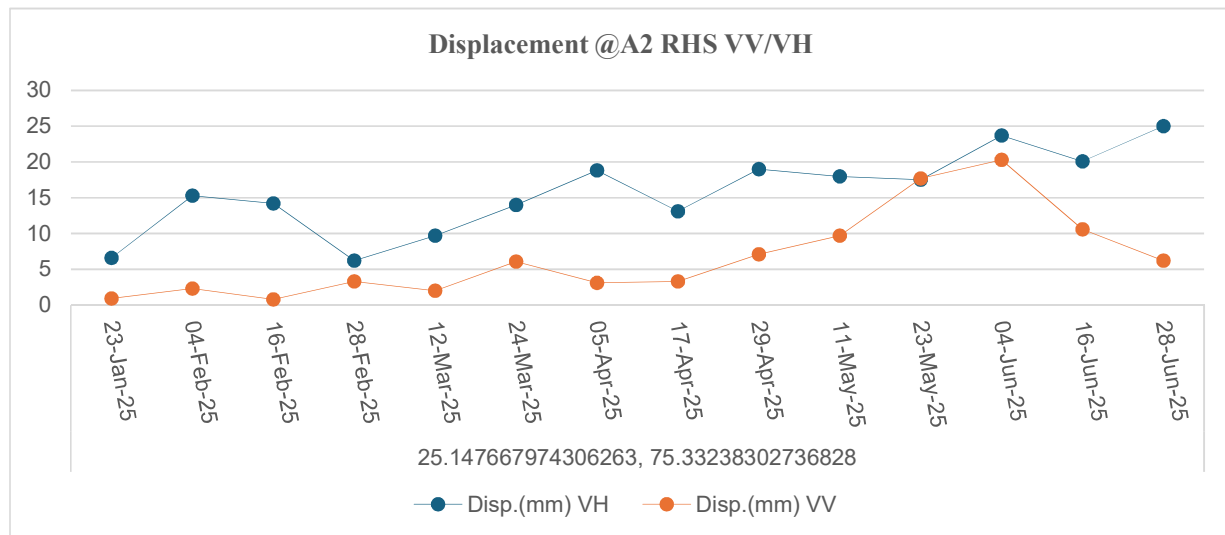
Critical Events & Anomalies:

- **January-February 2025:** VH shows moderate negative values (-5 to -10 mm) while VV remains near neutral to slightly negative, indicating initial lateral dominance with minimal vertical response.
- **June 2025:** VH again trends more negative (up to -17 mm) while VV remains consistently positive (+6 to +9 mm), suggesting persistent lateral movement under sustained uplift conditions.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.147667974306263, 75.33238302736828)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV is positive throughout, starting at low magnitudes (+1–2 mm) and progressively increasing to a peak of approximately +20 mm in early June.
- **Horizontal (VH) displacement** VH remains consistently positive and high across the monitoring period, increasing overall from approximately +6 mm in late January to +25 mm by late June 2025.

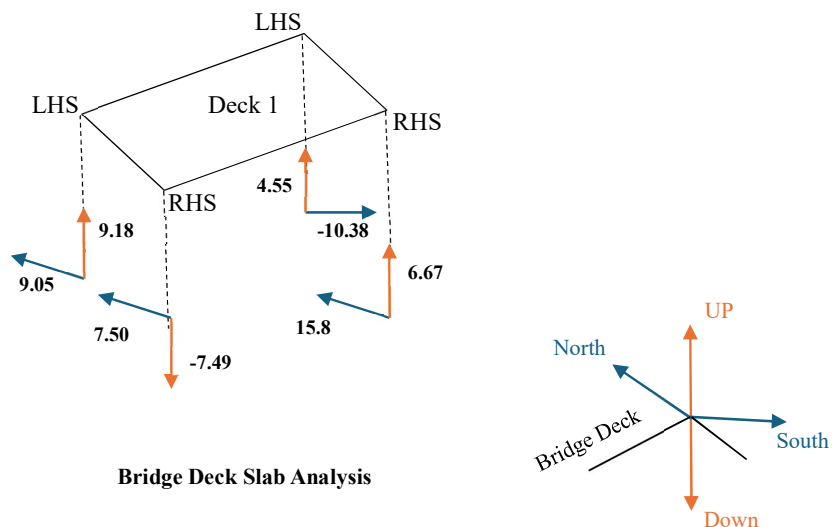
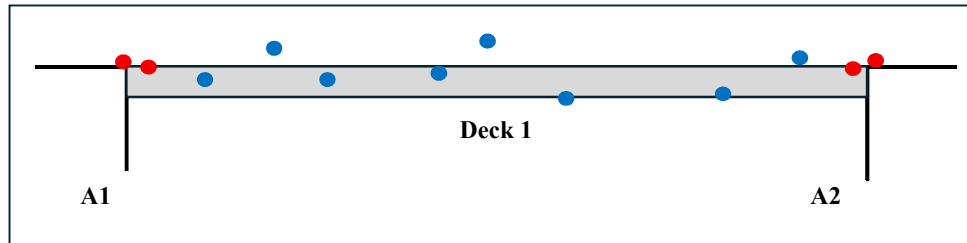
Critical Events & Anomalies:

- **January-February 2025:** VH shows strong positive values (+14–15 mm) while VV remains low (+1–3 mm), indicating lateral displacement dominance with minimal vertical response.
- **March 2025:** Both VH and VV increase gradually; VH reaches +14–19 mm, while VV rises to +5–6 mm, suggesting coupled lateral and vertical deformation.
- **June 2025:** VV reaches its maximum (+20 mm) as VH also peaks (+23 mm), representing a period of combined maximum deformation. VV decreases to +6–10 mm while VH continues to increase, peaking near +25 mm, indicating renewed lateral dominance after vertical relaxation.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab shows a non-uniform vertical displacement pattern across LHS and RHS supports. A moderate uplift of approximately +9.18 mm is observed at the LHS lower support, indicating upward movement at this location. The deck slab level near RHS records an uplift of approximately +4.55 mm, suggesting localized upward flexure of the slab.

Horizontal Displacement Behaviour:

- Horizontal displacement magnitudes reveal clear directional asymmetry between LHS and RHS regions. The LHS upper region shows a horizontal displacement of approximately +9.05 mm, indicating notable lateral movement towards the indicated direction. The RHS outer support records the maximum horizontal displacement of approximately +15.8 mm.

CONCLUSION:

- The bridge deck slab exhibits a distinctly non-symmetric displacement response, governed by combined vertical bending and torsional effects. Long-term observation is recommended to determine whether the observed behaviour is load-induced, thermal in nature, or indicative of evolving boundary condition effects.